Activity	1-2	1-3	2-4	2-5	3-4	4-5
Duration (in days)	8	4	10	2	5	3

Solution:

Earliest start time (EST) and latest finish time (LFT) of each activity are given ithe following network.

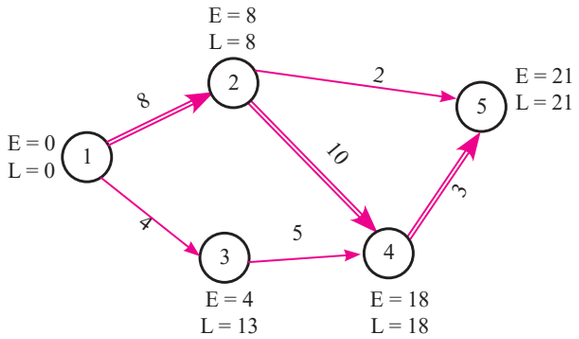


Fig 10.9

$E_1 = 0$

$E_2 = E_1 + t_{12} = 0 + 8 = 8$

$E_3 = E_1 + t_{13} = 0 + 4 = 4$

$E_4 = E_2 + t_{24}$ or $E_3 + t_{34}$
$= 8 + 10 = 18$
(take $E_2 + t_{24}$ or $E_3 + t_{34}$ whichever is maximum)

$E_5 = (E_2 + t_{25}$ or $E_4 + t_{45})$
$= 18 + 3 = 21$
(take $E_2 + t_{25}$ or $E_4 + t_{45}$ whichever is maximum)

$L_5 = 21$

$L_4 = L_5 - t_{45} = 21 - 3 = 18$

$L_3 = L_4 - t_{34} = 18 - 5 = 13$

$L_2 = L_5 - t_{25}$ or $L_4 - t_{24}$
$= 18 - 10 = 8$
(take $L_5 - t_{25}$ or $L_4 - t_{24}$ whichever is minimum)

$L_1 = L_2 - t_{12}$ or $L_3 - t_{13}$
$= 8 - 8 = 0$
(take $L_2 - t_{12}$ or $L_3 - t_{13}$ whichever is minimum)

Here the critical path is 1-2-4-5, which is denoted by double lines.

Activity	Duration (t_{ij})	EST	EFT=EST+t_{ij}	LST=LFT-t_{ij}	LFT
1–2	8	0	8	0	8
1–3	4	0	4	9	13
2–4	10	8	18	8	18
2–5	2	8	10	19	21
3–4	5	4	9	13	18
4–5	3	18	21	18	21

Table 10.5

The longest duration to complete this project is 21 days.

The path connected by the critical activities is the critical path (the longest path).

Critical path is 1-2-4-5 and project completion time is 21 days.

Example 10.14

Calculate the earliest start time, earliest finish time, latest start time and latest finish time of each activity of the project given below and determine the Critical path of the project and duration to complete the project.

Activity	1-2	1-3	1-5	2-3	2-4	3-4	3-5	3-6	4-6	5-6
Duration (in week)	8	7	12	4	10	3	5	10	7	4

Solution:

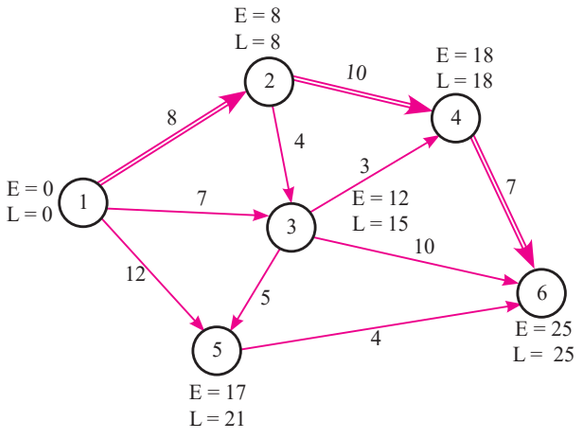


Fig 10.10

Activity	Duration (in week)	EST	EFT	LST	LFT
1-2	8	0	8	0	8
1-3	7	0	7	8	15
1-5	12	0	12	9	21
2-3	4	8	12	11	15
2-4	10	8	18	8	18
3-4	3	12	15	15	18
3-5	5	12	17	16	21
3-6	10	12	22	15	25
4-6	7	18	25	18	25
5-6	4	17	21	21	25

Table 10.6

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